

CS & IT ENGINEERING



THEORY OF COMPUTATION

DFA

Lecture No.- 08



By- Venkat sir



Recap of Previous Lecture



Topic

??????

DFA Difficult for some languages

NFA Construction is easy

$NFA \Rightarrow DFA$



Topics to be Covered



ϵ -NFA



Topic

Conversion from ϵ -NFA to NFA

Topic

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ϵ -NFA $\Rightarrow$ DFA

Topic

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Topic

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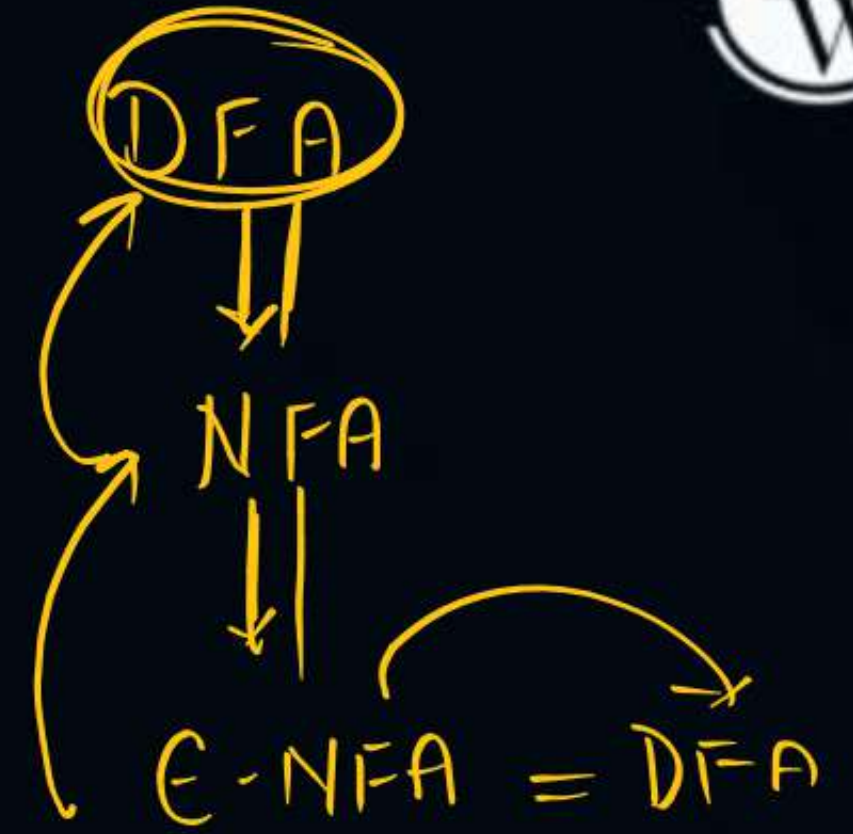
Topic : ϵ -NFA

NOTE: Construction of ϵ - NFA is easy than NFA

$\{Q, \Sigma, q_0, F, \delta\}$

Q	-	Finite number of states (set of state)
Σ	-	Input alphabet
q_0	-	initial state
F	-	Set of final states
δ	-	transition function

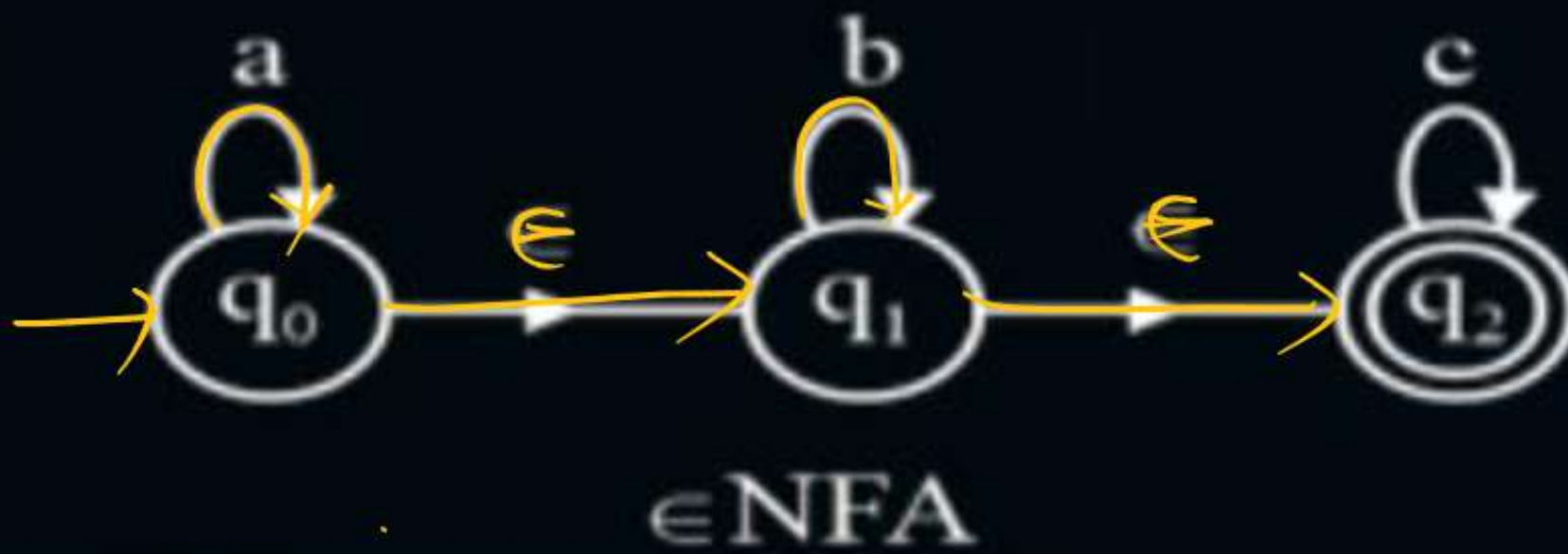
$$\delta: Q \times \Sigma \cup \{\epsilon\} \rightarrow 2^Q$$





Topic : ϵ -NFA

$L = \{a^n b^m c^k / n, m, k \geq 0\}$ construct ϵ -NFA for L





Topic : ϵ -NFA

While converting ϵ -NFA into NFA (without ϵ) the following are the possibilities

- ① No. of states are same
- ② Initial state is same
- ③ Final state may changes (may increase)
- ④ Transitions may changes



Topic : Conversion from ϵ -NFA to NFA

1. Number of states in ϵ -NFA is same of NFA
2. Initial state of ϵ - NFA is same as NFA
3. In NFA make states as final where ϵ -closure of that state contains a final state of ϵ -NFA.



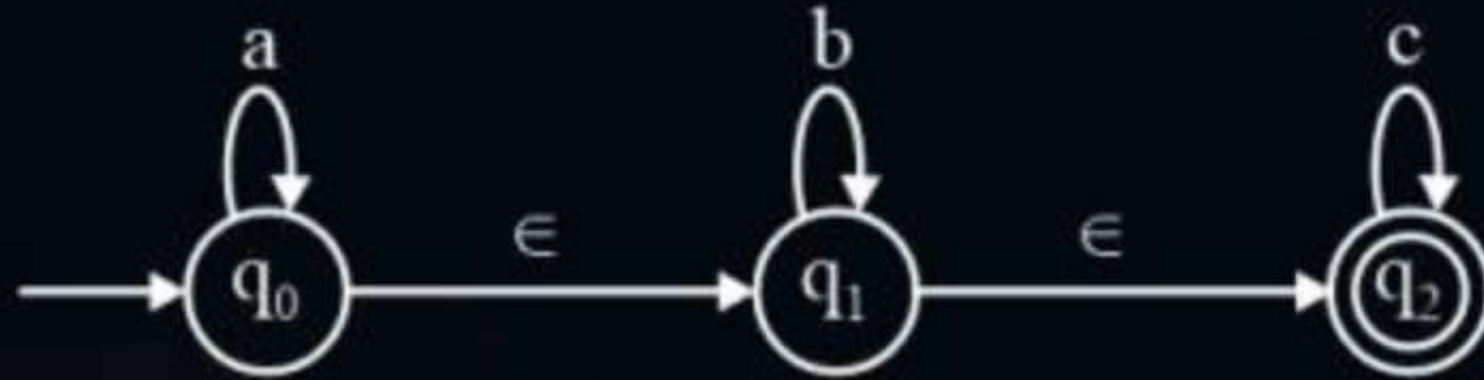
Topic : Conversion from ϵ -NFA to NFA

ϵ -closure (q) = set of all states which are reachable from state q by reading only ϵ .

#Q. Construct an equivalent NFA for the following ϵ -NFA

$\epsilon \cdot \epsilon \cdot \epsilon = \epsilon$

ϵ -NFA



NFA



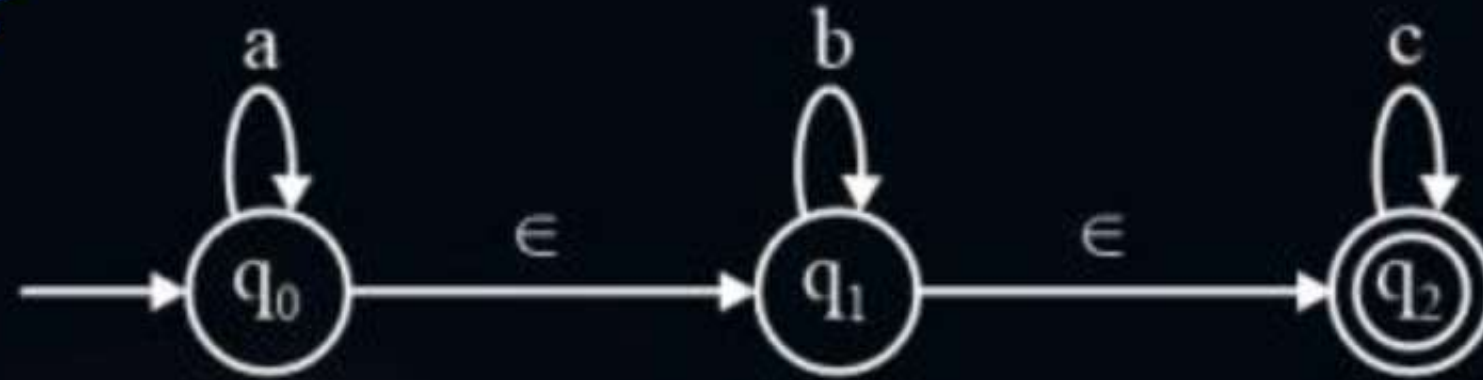
$$\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure}(q_1) = \{q_1, q_2\}$$

$$\epsilon\text{-closure}(q_2) = \{q_2\}$$

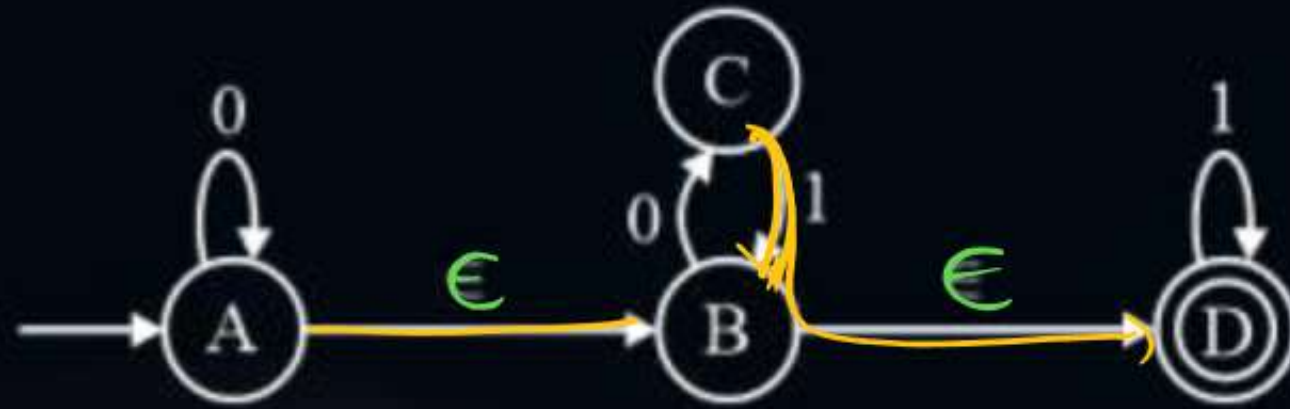
#Q. Construct an equivalent NFA for the following ϵ -NFA

Home Work



#Q. Construct an equivalent NFA for the following ϵ -NFA

ϵ -NFA

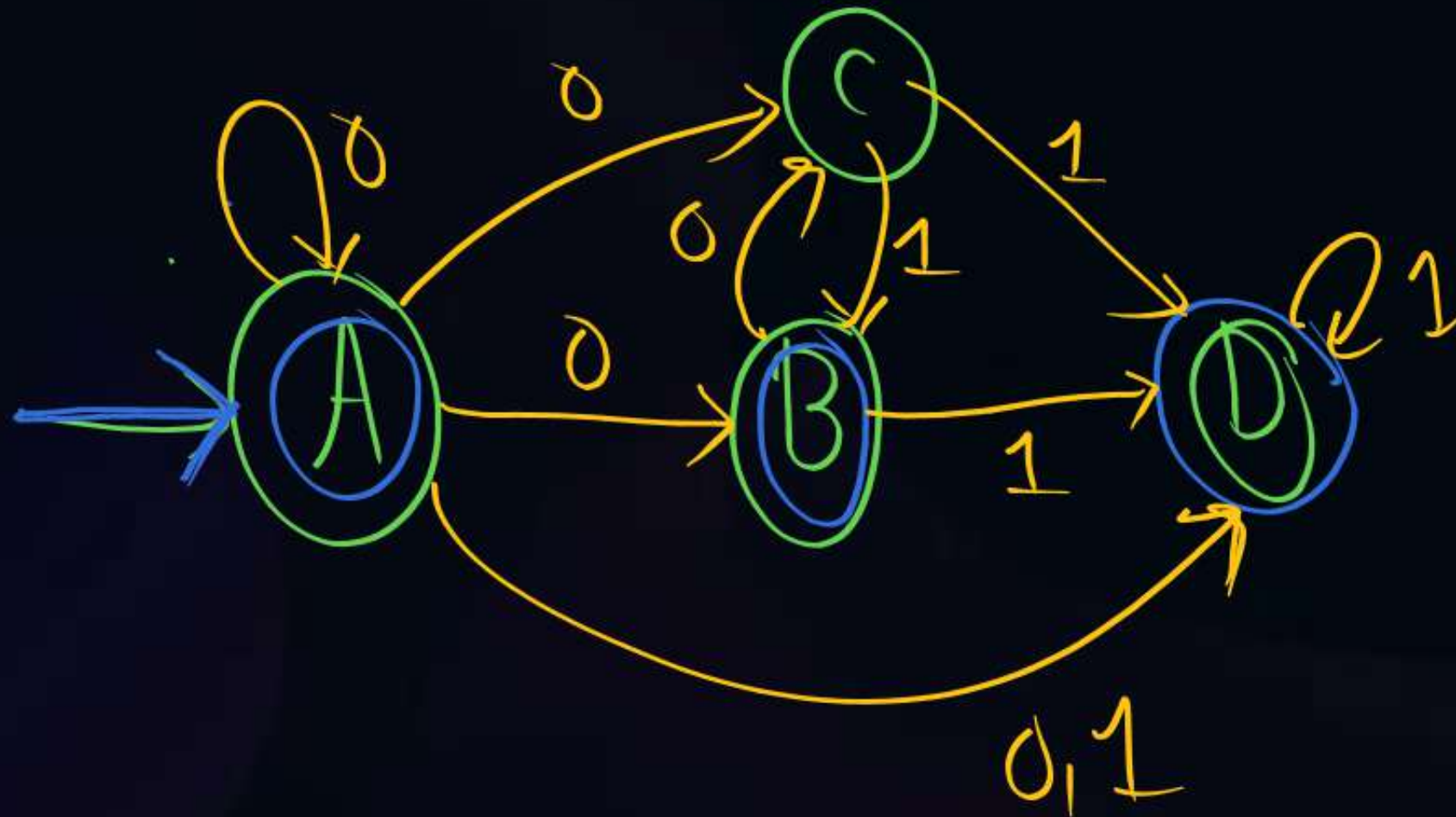


$$\epsilon\text{-closure}(A) = \{A, B, D\}$$

$$\epsilon\text{-closure}(B) = \{B, D\}$$

$$\underline{S(A, 0)} = \{A, B, C, D\}$$

NFA



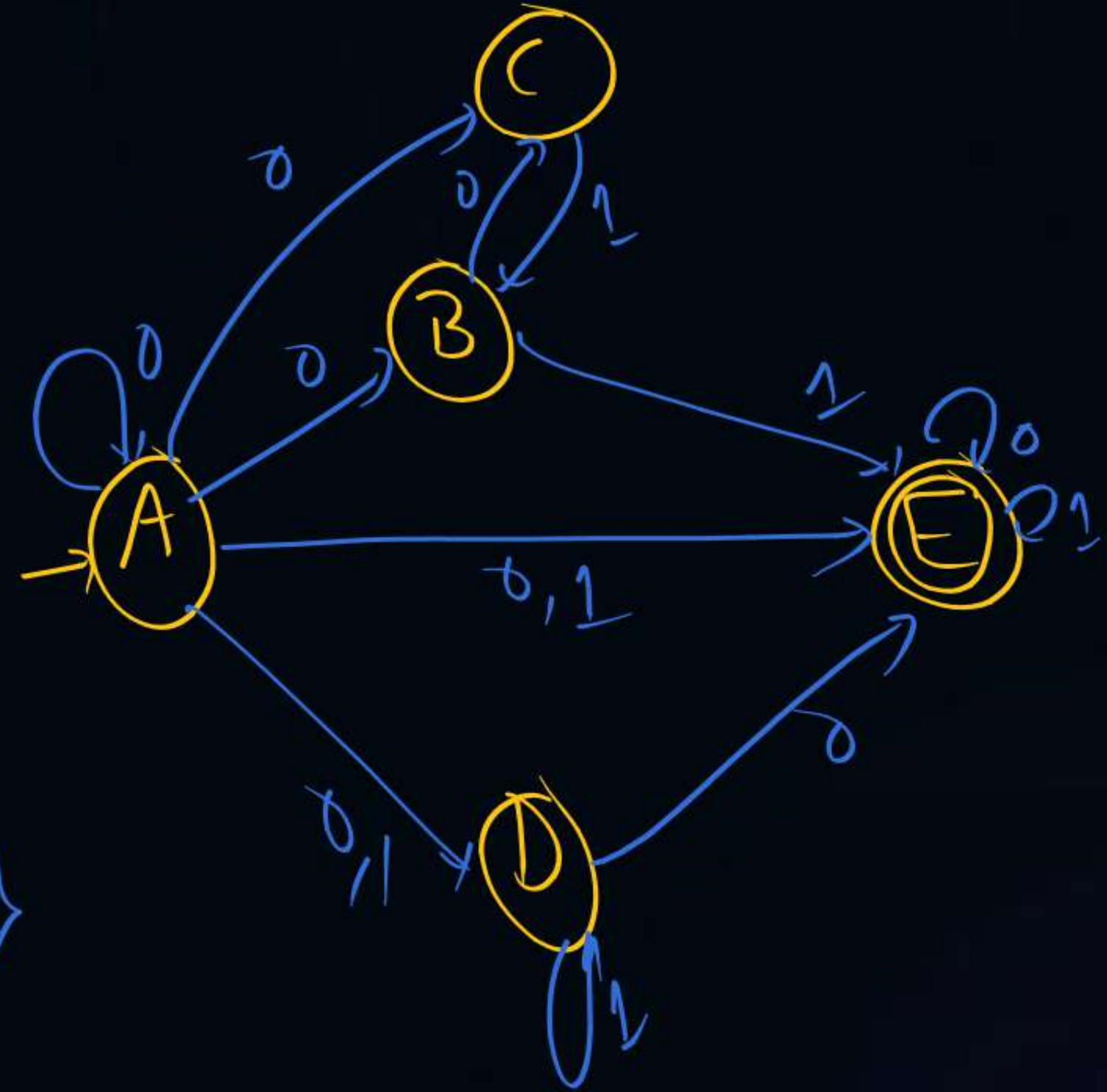
#Q. Construct an equivalent NFA for the following ϵ -NFA

③



$$\delta(A, 0) = \{A, B, D, E\}$$

$$\delta(A, 1) = \{D, E\}$$



#Q. Construct an equivalent NFA for the following ϵ -NFA



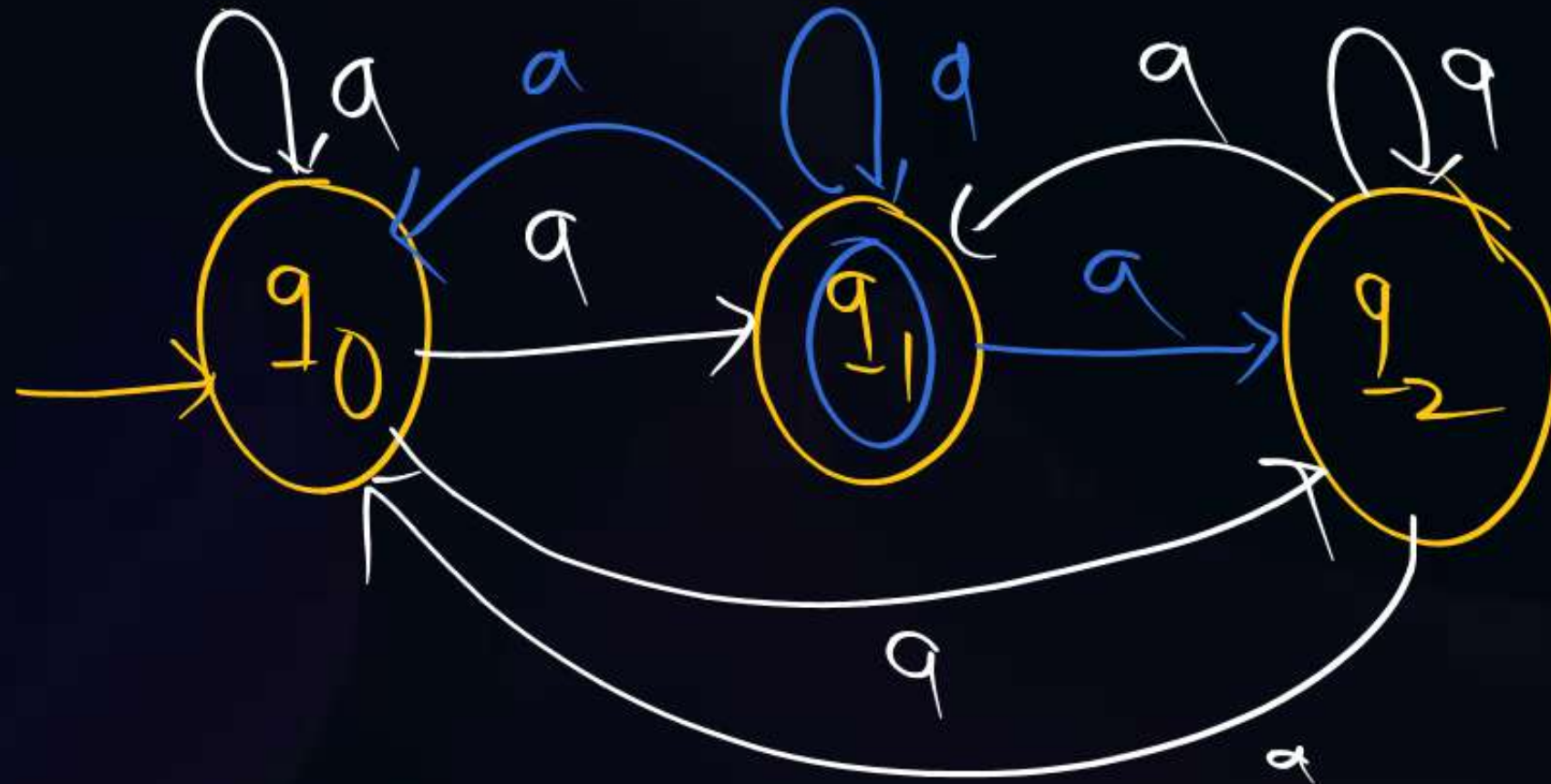
#Q. Construct an equivalent NFA for the following ϵ -NFA

$\delta(q_0, a)$
 ① method

ϵ NFA



NFA



$$\epsilon\text{-closure}(q_0) = \{q_0\}$$

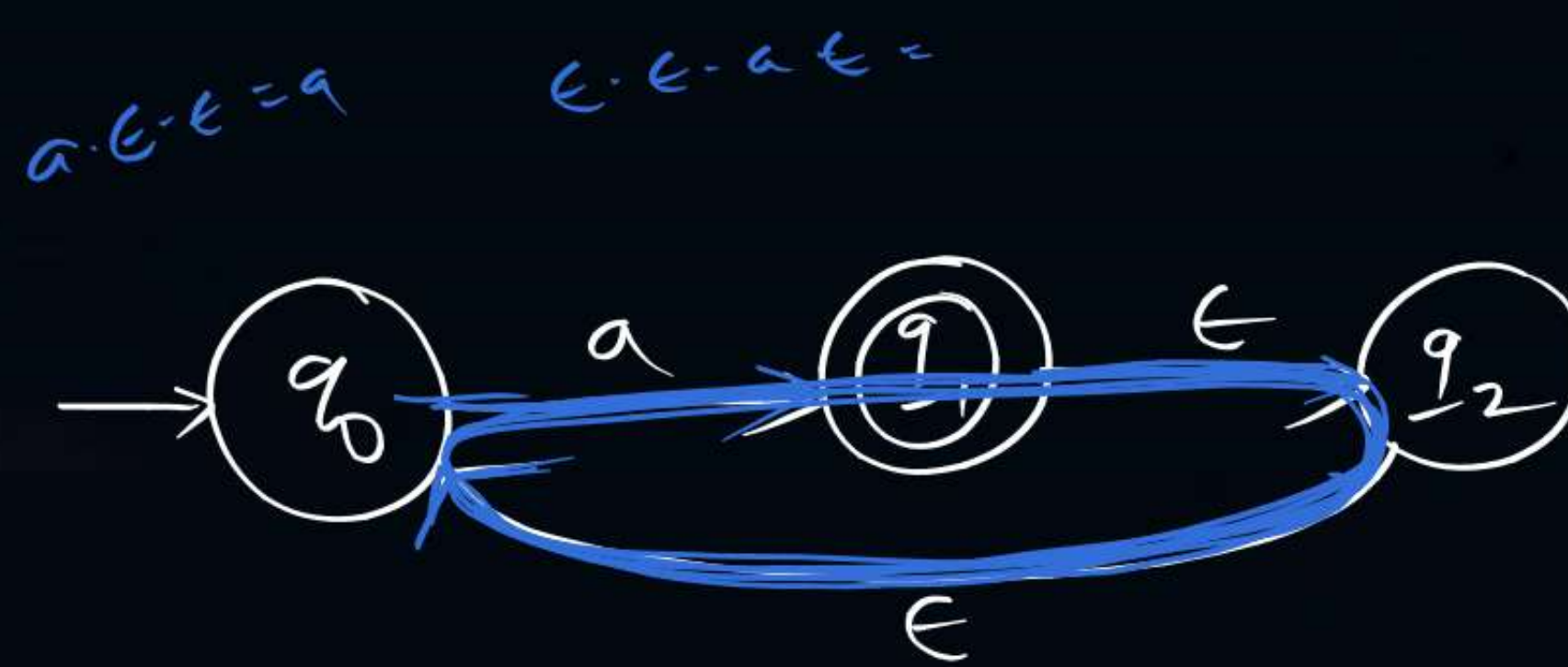
$$\epsilon\text{-closure}(q_1) = \{q_1, q_2, q_0\}$$

$$\epsilon\text{-closure}(q_2) = \{q_0, q_2\}$$

$$\delta(q_0, a) = \{q_0, q_1, q_2\}$$

$$\delta(q_1, a) = \{q_0, q_1, q_2\}$$

$$\delta(q_2, a) = \{q_0, q_1, q_2\}$$



$$\left[\begin{array}{l} \epsilon \cdot \epsilon \cdot \epsilon = \epsilon \\ \epsilon \cdot \epsilon \cdot a \cdot \epsilon = a \end{array} \right]$$





Topic : Conversion from ϵ -NFA to NFA

$$\delta(q_2, a) = \epsilon\text{-closure} \left[\delta(\epsilon\text{-closure}(q_2), a) \right]$$

Transitions of NFA is $\left[\begin{matrix} \delta(q_2, a) \\ \emptyset \end{matrix} \right] \cup \left[\begin{matrix} \delta(q_0, a) \\ q_1 \end{matrix} \right] \Rightarrow \epsilon\text{-closure}(q_1) = \{q_0, q_1, q_2\}$

$$\delta^1(q_1, a) = \epsilon\text{-closure} \left[\delta(\epsilon\text{-closure}(q_1), a) \right]$$

$$\delta(q_1, a) = \epsilon\text{-closure} \left[\delta(\epsilon\text{-closure}(q_0), a) \right]$$

$$\Downarrow$$
$$\{q_0, q_1, q_2\}$$

$$\epsilon\text{-closure} \left[\delta(q_0, a) \right]$$

$$\epsilon\text{-closure}(q_1) = \{q_0, q_1, q_2\}$$



Topic : Conversion from ϵ -NFA to NFA

Transitions of NFA is

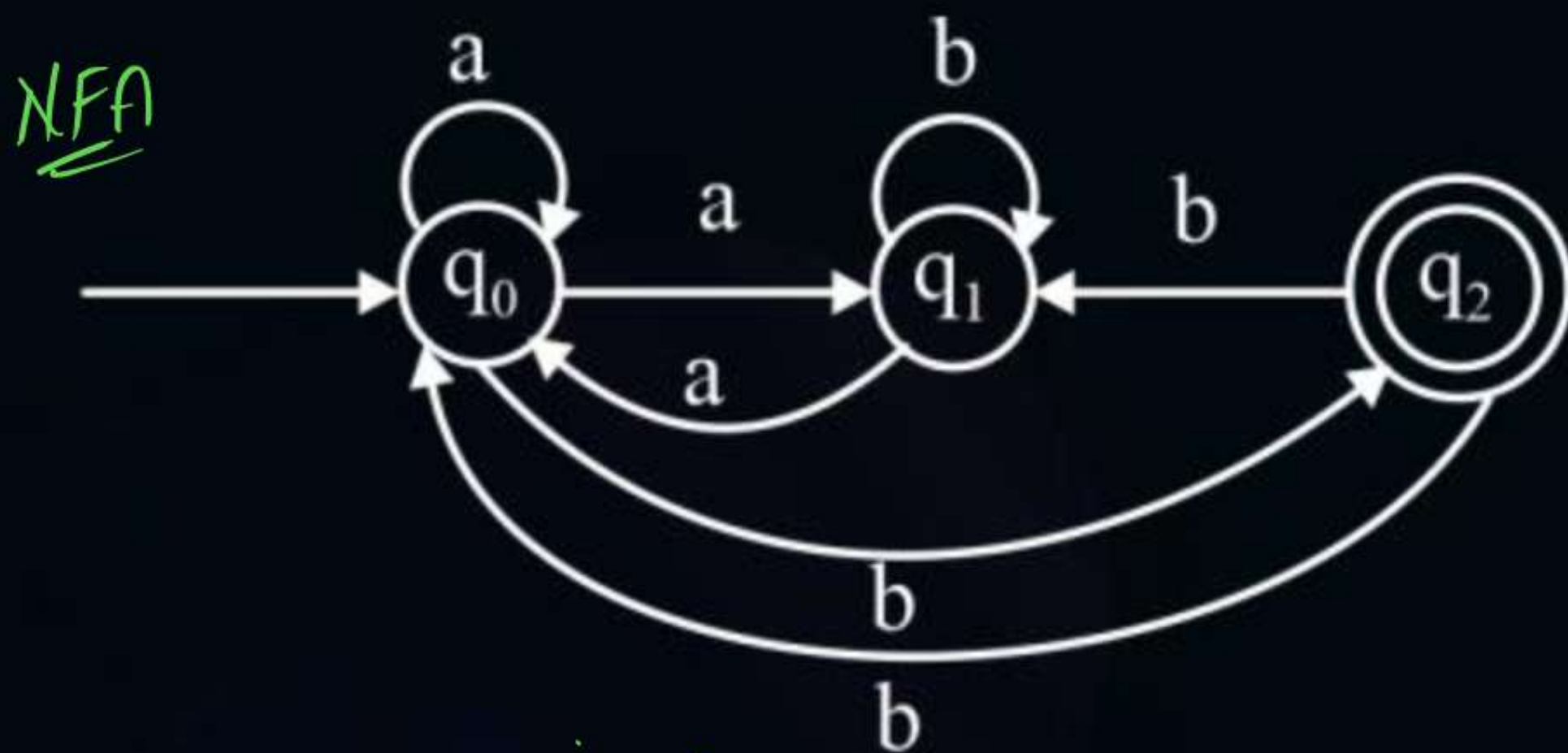
$$\delta^1(q_1, a) = \underline{\epsilon\text{-closure}}(\delta(\epsilon\text{-closure}(q), a))$$

$$\begin{aligned} \delta(q_1, a) &= \left[\delta[q_0, q_1, q_2], a \right] \\ &\Downarrow \\ \{q_0, q_1, q_2\} &\quad \underline{\epsilon\text{-closure}} \left[(q_0, a) \cup (q_1, a) \cup (q_2, a) \right] \\ &\quad \quad \quad \downarrow \quad \downarrow \quad \downarrow \\ &\quad \quad \quad q_1 \cup \emptyset \cup \emptyset \\ &\quad \quad \quad \epsilon\text{-closure}[q_1] = \{q_0, q_1, q_2\} \end{aligned}$$



Topic : NFA to DFA Conversion

Consider the following NFA what is the equivalent DFA for given NFA?



	a	b
→ q ₀	{q ₀ , q ₁ }	q ₂
q ₁	q ₀	q ₁
(q ₂)	-	{q ₀ , q ₁ }

DFA

	a	b
→ q ₀	[q ₀ , q ₁]	q ₂
[q ₀ , q ₁]	[q ₀ , q ₁]	[q ₀ , q ₁ , q ₂]
(q ₂)	q ₀	[q ₀ , q ₁]
[q ₀ , q ₁ , q ₂]	q ₀	[q ₀ , q ₁]
q ₀	q ₀	q ₁

Sub set Construction

$$Q \times \Sigma \rightarrow \underline{\underline{2^Q}}$$

NFA $\Rightarrow$ DFA

$$\textcircled{1} \quad 2 \longrightarrow 4$$

$$\textcircled{2} \quad 3 \longrightarrow 5$$

$$3 \longrightarrow 4$$

$$\underline{\underline{\eta_{\text{state}}}} \longrightarrow \left\{ \underline{\underline{\text{min} \rightarrow 1 \sim \text{max} \rightarrow 2^n}} \right\}$$

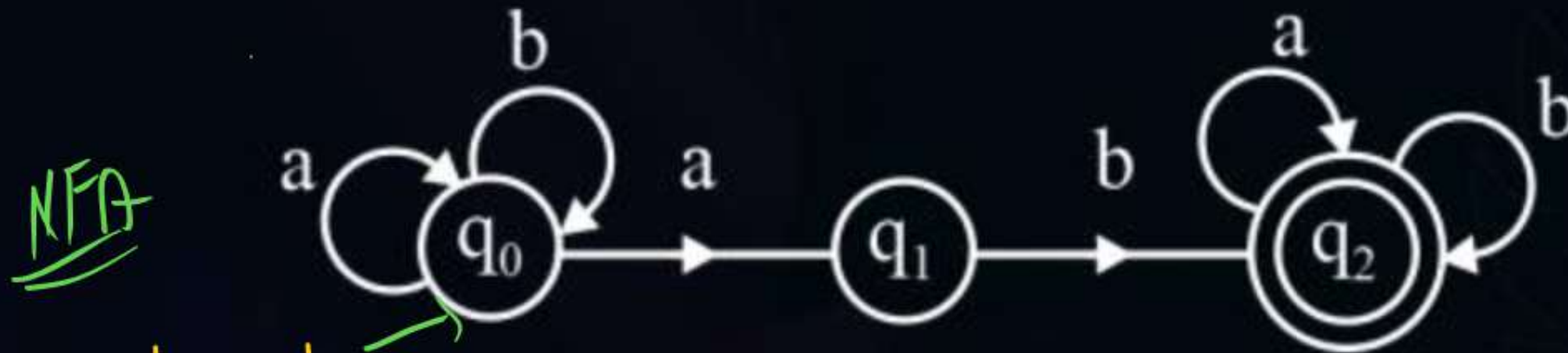


Topic : NFA to DFA Conversion

Note:- NFA to DFA conversion Algorithm does not affect language of automata i.e. if language accept by a NFA is L then while converting that NFA into DFA the resultant DFA also accepts.

NFA $\rightarrow$ DFA min DFA ? ③

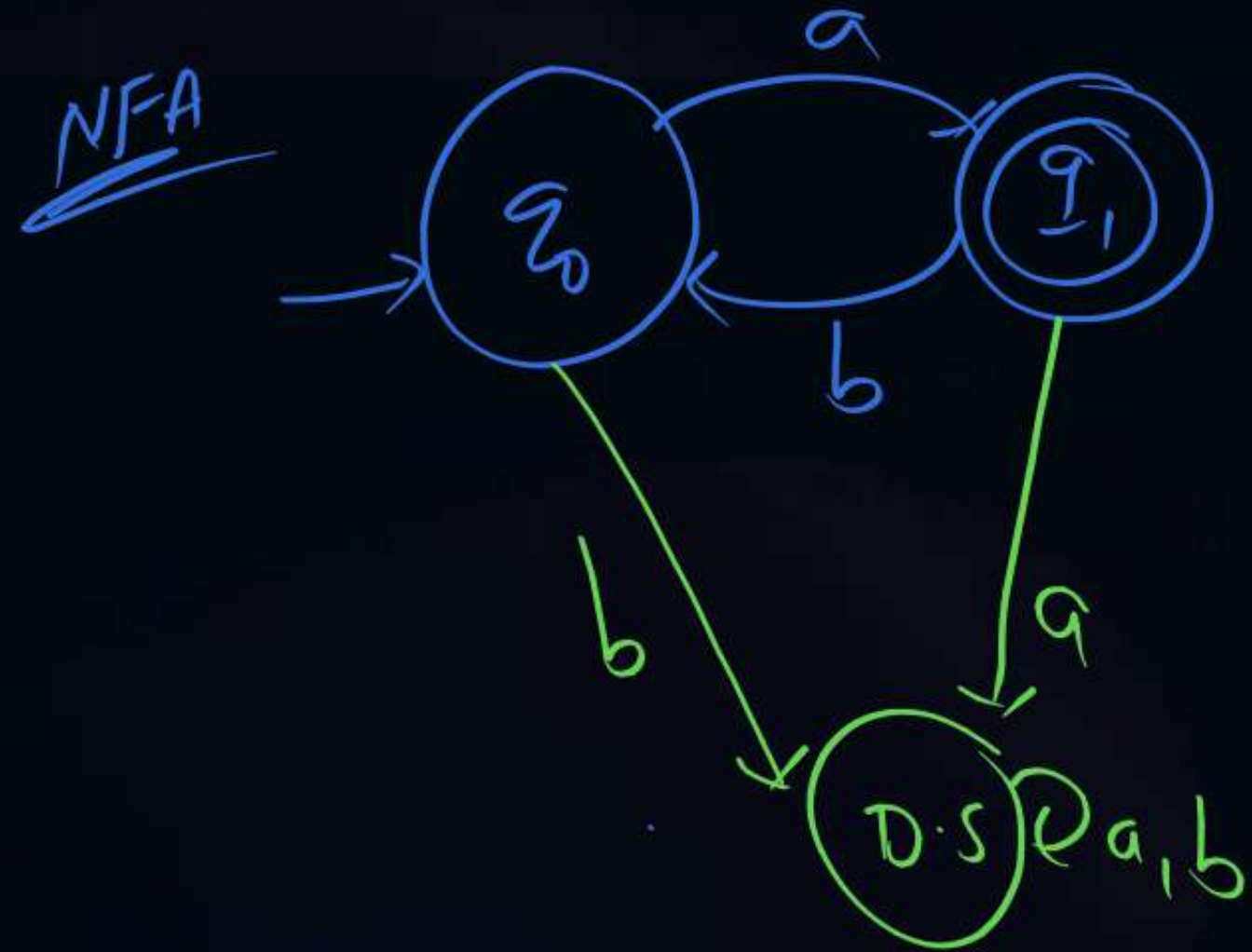
NFA: min DFA



	a	b
q ₀	{q ₀ , q ₁ }	q ₀
q ₁	-	q ₂
q ₂	q ₂	q ₂

	a	b
→ q ₀	[q ₀ , q ₁]	q ₀
[q ₀ , q ₁]	[q ₀ , q ₁]	[q ₀ , q ₂]
[q ₀ , q ₂]	[q ₀ , q ₁ , q ₂]	[q ₀ , q ₂]
[q ₀ , q ₁ , q ₂]	[q ₀ , q ₁ , q ₂]	[q ₀ , q ₂]

(Q) Construct DFA for given NFA

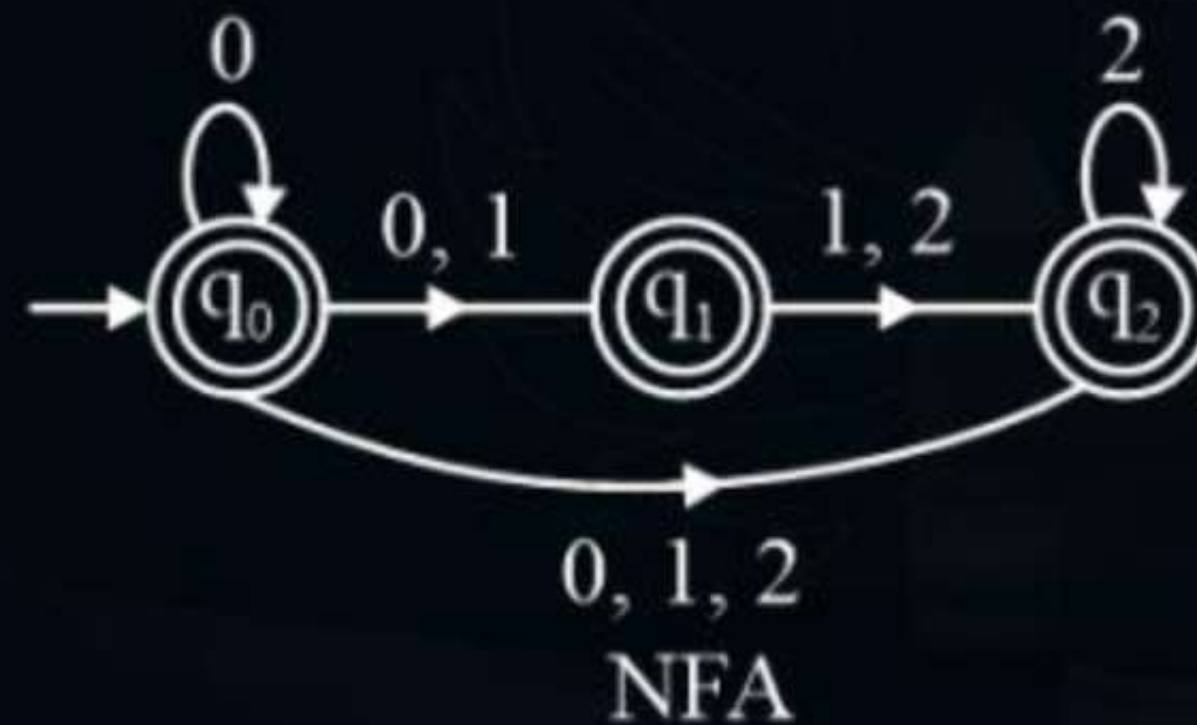
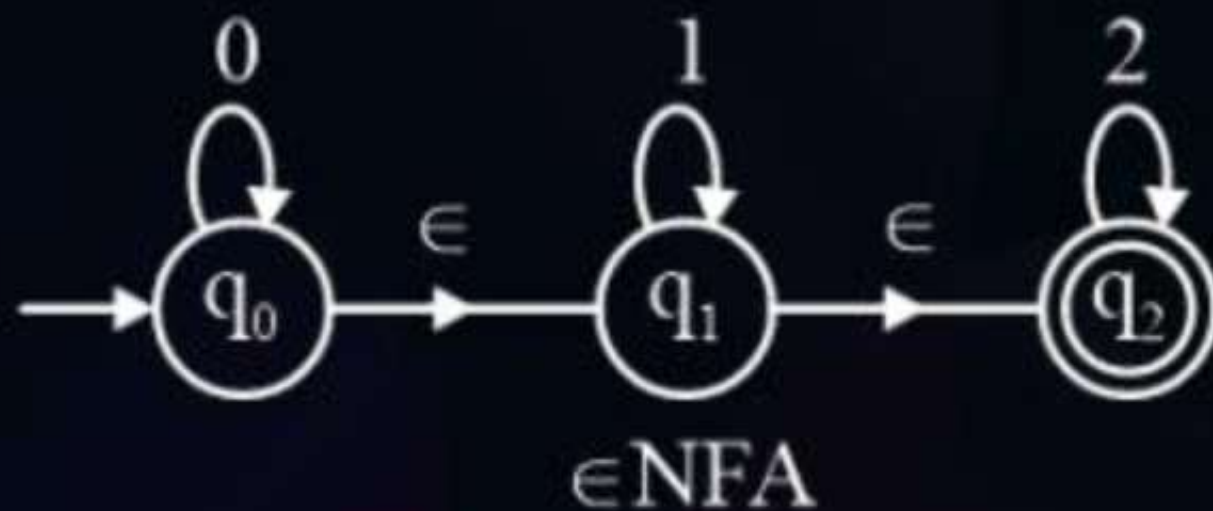




Topic : ϵ -NFA

While converting ϵ -NFA into NFA (with ϵ) the following act the possibility

- No. of states are same
- Initial state is same
- Final state may changes
- Transition may changes



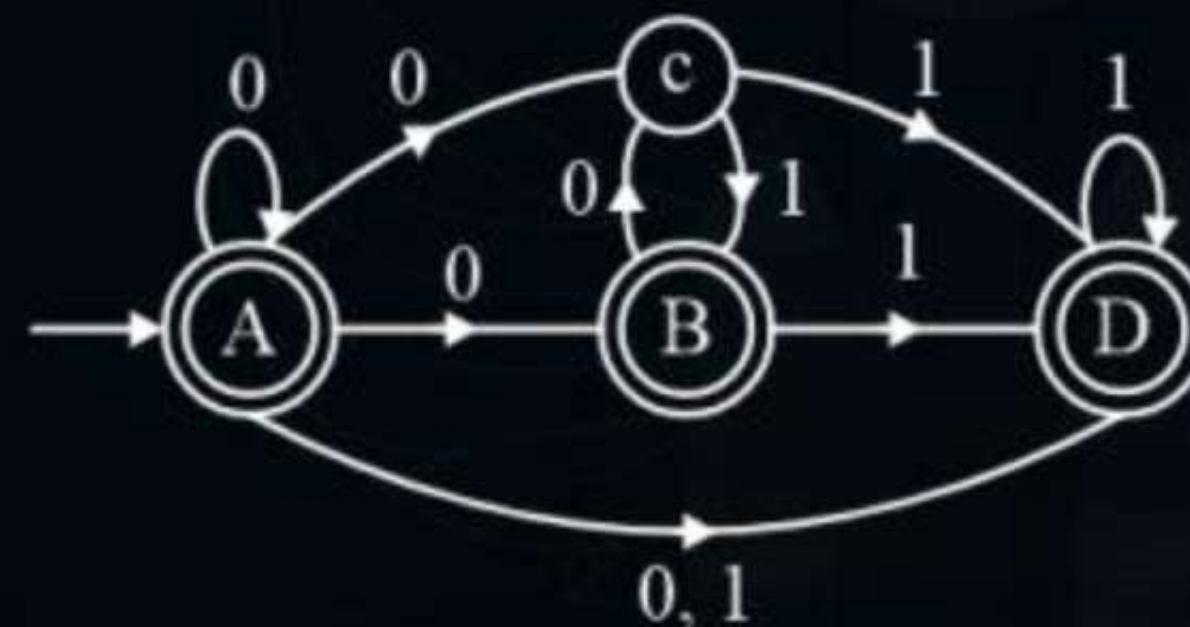
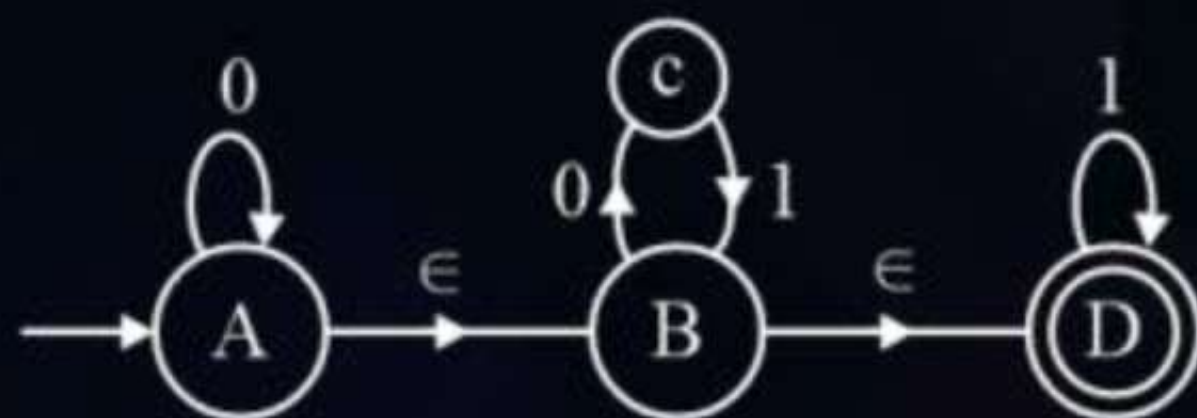


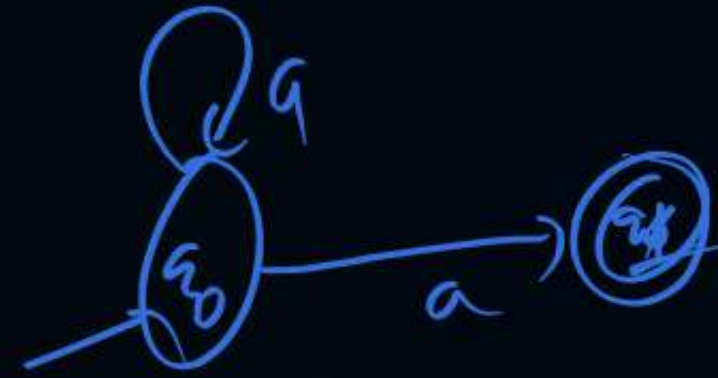
Topic : ϵ -NFA

In ϵ -NFA from any state by reading only ϵ if final state is reachable then the state becomes final state in NFA without ϵ .

Basics:

$1.\epsilon = 1$: $0.\epsilon = 0$: $\epsilon.0.\epsilon = 0$, $\epsilon.\epsilon = \epsilon$





a

$$\underline{\delta(q_0, a) = q_0} \quad \times$$

THANK - YOU

$$\underline{\delta(q_0, a) = q_1} \quad \checkmark$$